Supply Chain Optimization

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Course book: “Inventory control”
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SINGLE-ECHELON SYSTEMS

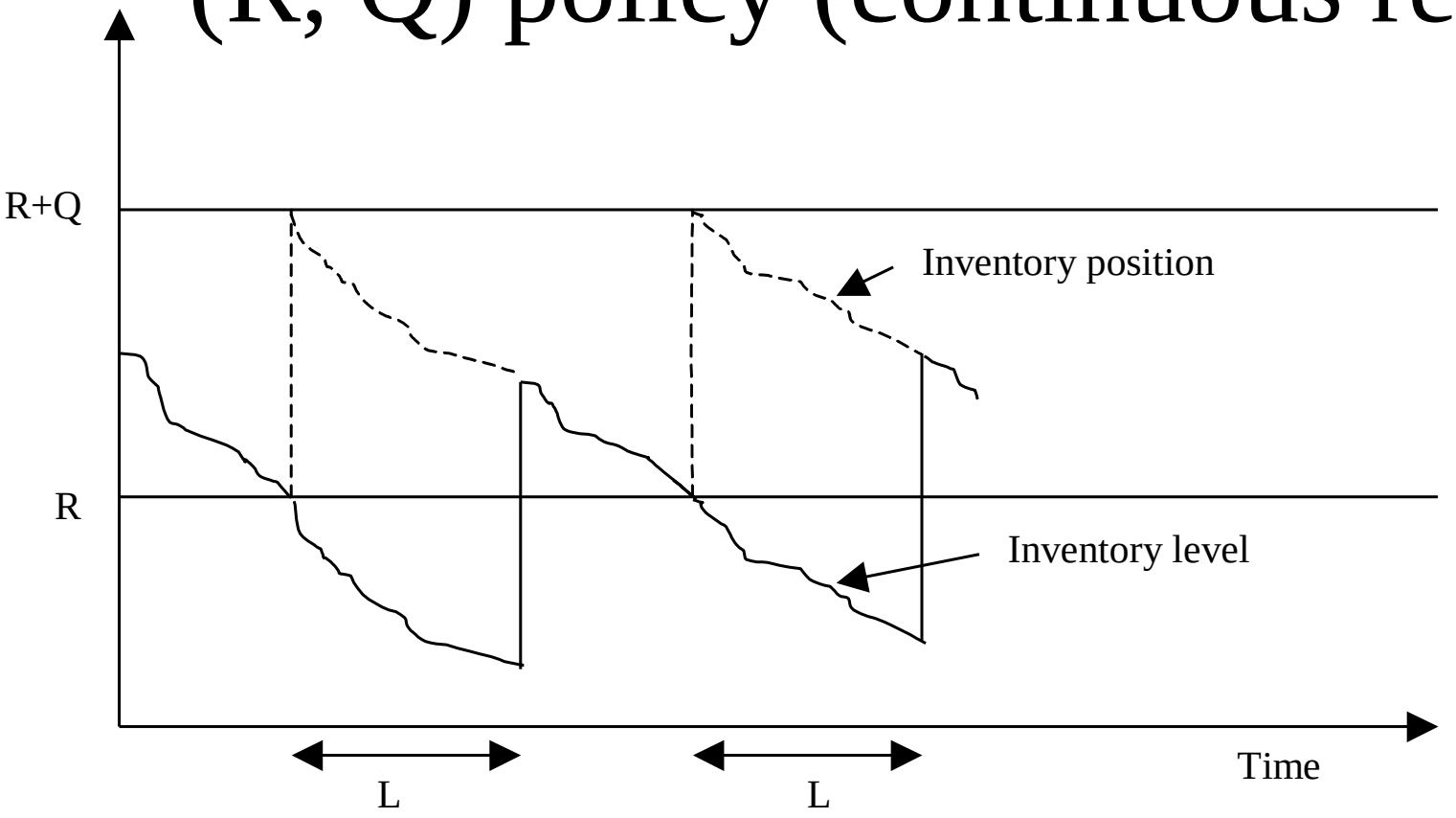
SINGLE ITEM

- Different items can be controlled independently.
- The items are stocked at a single location, i.e., not in a multi-stage inventory system.

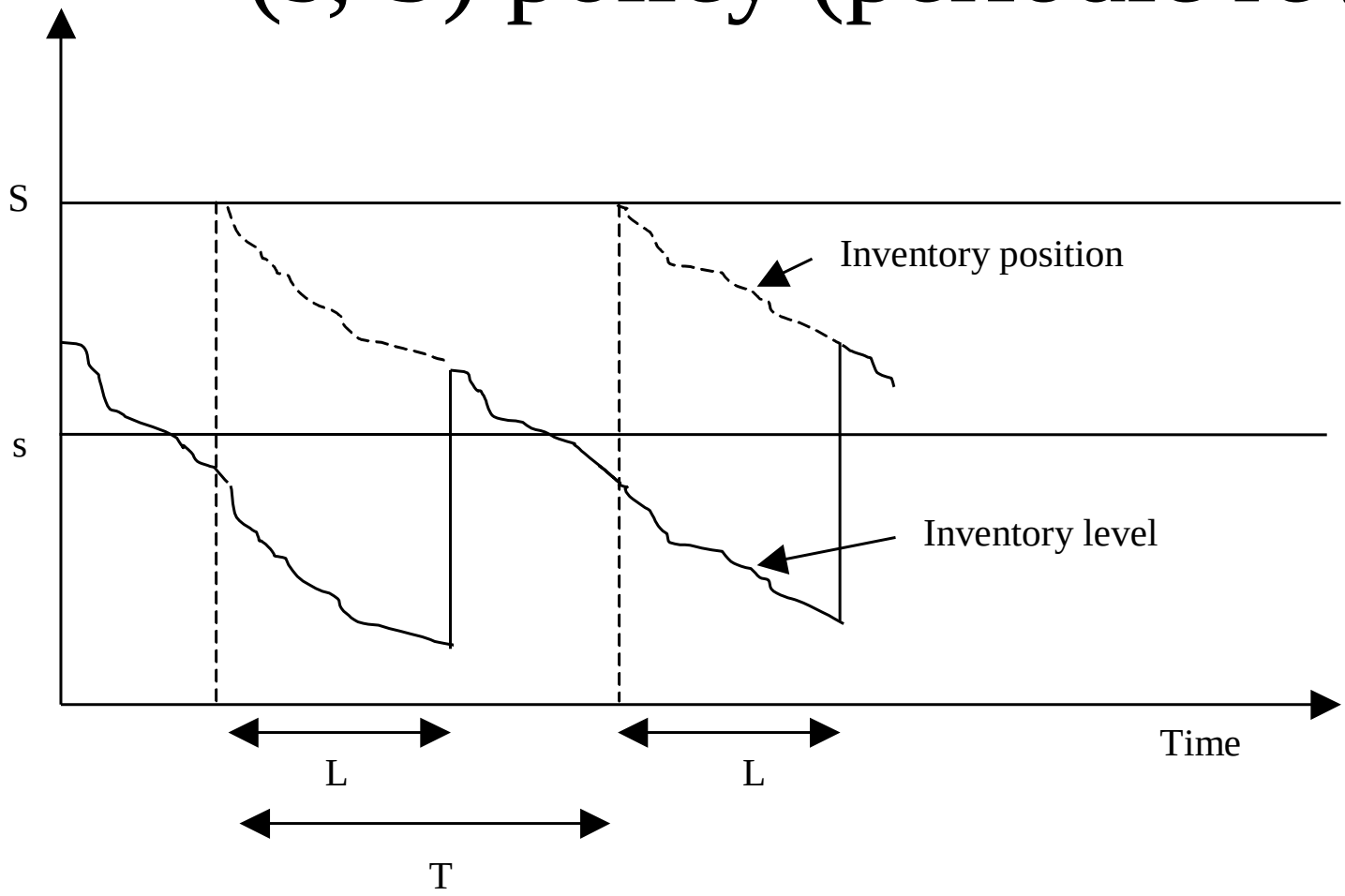
Different ordering policies

- Periodic review
- Continuous review

(R, Q) policy (continuous review)



(s, S) policy (periodic review)



Inventory definitions

- inventory position = stock on hand + outstanding orders - backorders
- inventory level = stock on hand - backorders

Goal

Determine the optimal order level (R or s) and the optimal order quantity (Q or S-s)

Costs, objectives, assumption

- Holding cost (per item per time unit)
- Ordering or set-up cost (per order)
- Shortage costs or service constraint

- Cost-objective: minimize holding, ordering, and shortage cost (shortage cost per late order arrival, per backorder, per backorder per time unit)
- Service-objective: minimize holding and ordering cost under a service constraint

ASSUMPTION: For this course, we assume that all customers not served immediately are willing to wait, i.e. are backordered. So there are no lost sales.

Order quantity

The classical *economic order quantity model (EOQ).*
It was first derived by Harris (1913), but also Wilson (1934) is recognized in connection with this model.

The model is based on the following assumptions:

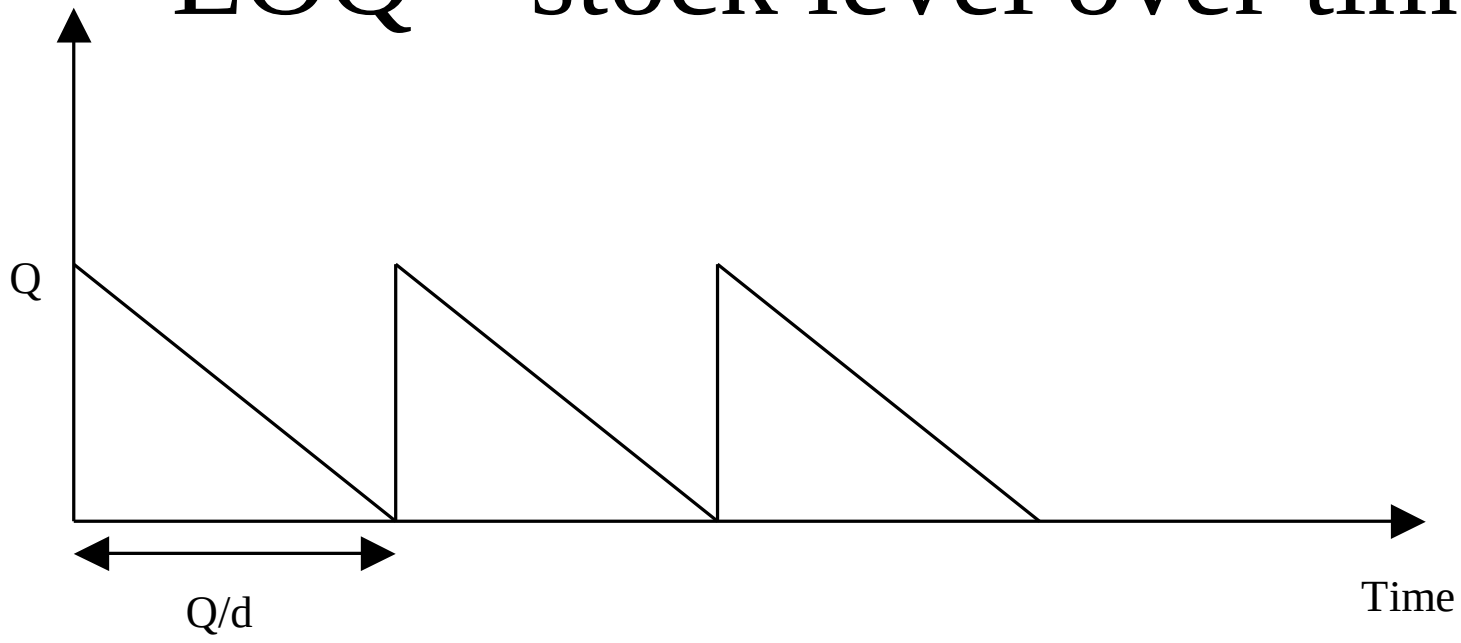
- Demand is constant and continuous.
- Ordering and holding costs are constant over time.
- The batch quantity does not need to be an integer.
- The whole batch quantity is delivered at the same time.
- No shortages are allowed.

EOQ

Notation:

h	=	holding cost per unit and time unit,
A	=	ordering or setup cost,
d	=	demand per time unit,
Q	=	batch quantity,
C	=	costs per time unit.

EOQ – stock level over time



EOQ (optimization)

Total cost $C = \frac{Q}{2}h + \frac{d}{Q}A$ is convex

$\frac{dC}{dQ} = \frac{h}{2} - \frac{d}{Q^2}A = 0$ gives $Q^* = \sqrt{\frac{2Ad}{h}}$ $C^* = \sqrt{\frac{Adh}{2}} + \sqrt{\frac{Adh}{2}} = \sqrt{2Adh}$

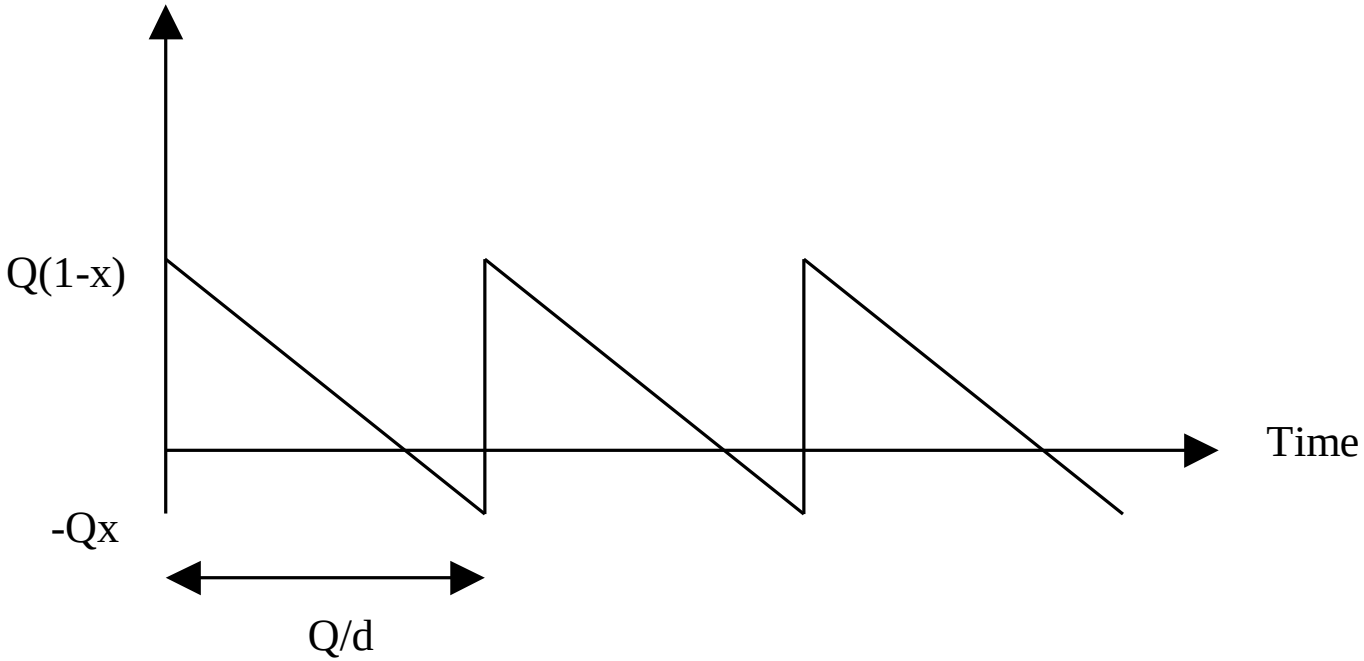
$$\frac{C}{C^*} = \frac{Q}{2}\sqrt{\frac{h}{2Ad}} + \frac{1}{2Q}\sqrt{\frac{2Ad}{h}} = \frac{1}{2}\left(\frac{Q}{Q^*} + \frac{Q^*}{Q}\right)$$

EOQ with backorders allowed

Let us define

b_1 = shortage penalty cost per unit and time unit.

x = fraction of demand that is backordered.



EOQ with backorders (continued)

$$C = \frac{Q(1-x)^2}{2} h + \frac{Qx^2}{2} b_1 + \frac{d}{Q} A$$

The cost is convex in x, the optimal x is independent of Q.

$$x^* = \frac{h}{h + b_1} \qquad C(x^*) = \frac{Q}{2} \frac{hb_1}{h + b_1} + \frac{d}{Q} A$$

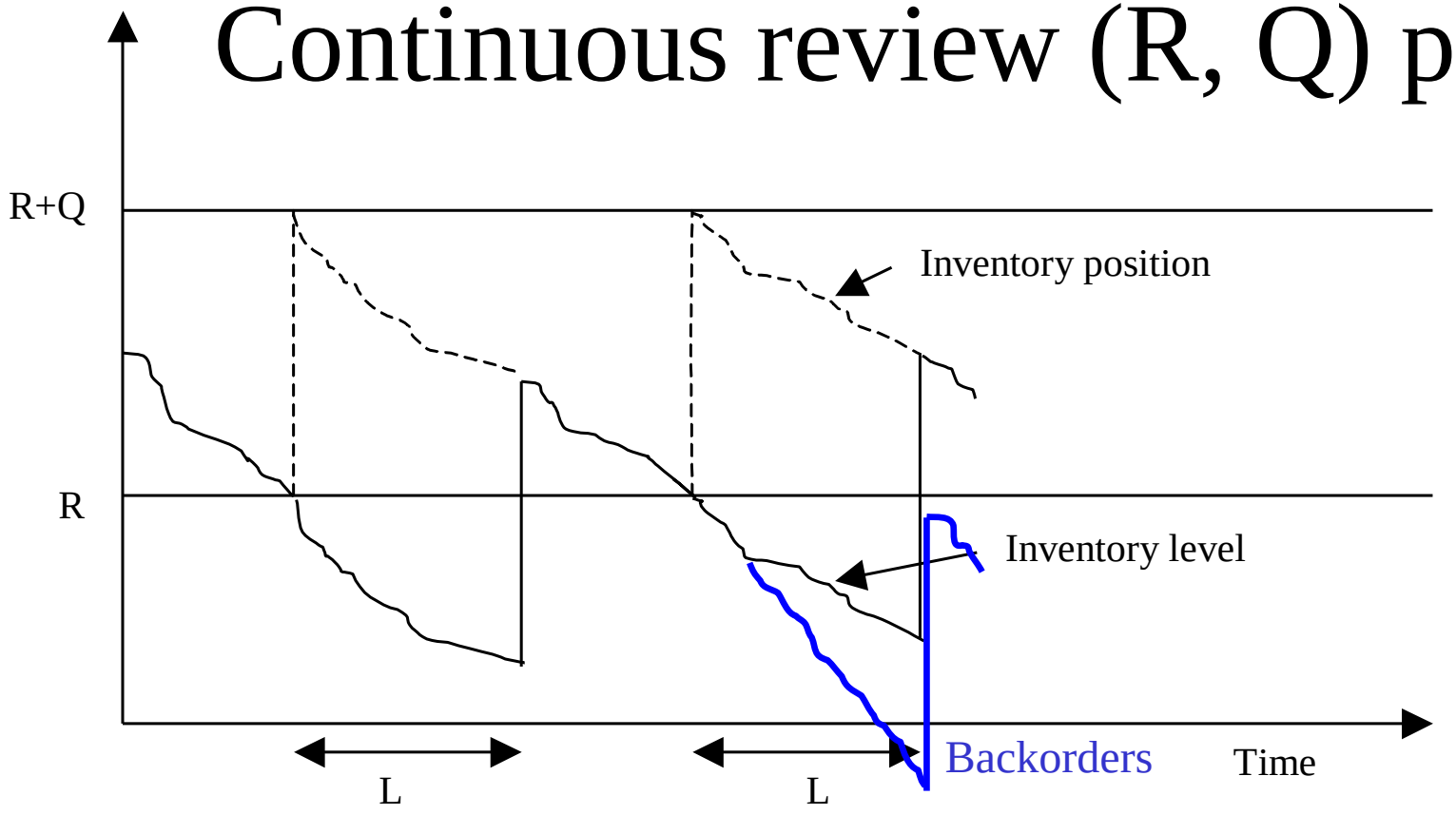
This gives

$$Q^* = \sqrt{\frac{2Ad\,(h + b_1)}{hb_1}} \qquad C^* = \sqrt{\frac{2Ad\,hb_1}{(h + b_1)}}$$

(classical economic lot size problem as $b_1 \to \infty$)

Safety stocks and reorder points

Continuous review (R, Q) policy



Lead time demand is crucial

Lead time demand distribution

- Poisson (pure or compounded)
- Normal

Standard Normal distribution

The standardized normal distribution with mean 0 and standard deviation 1 has density function

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

and distribution function

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

Normal distribution

The normal distribution with mean μ and standard deviation σ has density function

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(\frac{x-\mu}{\sigma}\right)^2}{2}}$$

FORECASTING: use point forecast for mean and MSE for variance of demand per time unit.

Poisson demand process

Poisson process with intensity λ:
in a small time interval Δt
the probability for exactly one customer is $\lambda\Delta t$,
the probability for more than one customer is $o(\Delta t)$.

The number of customers in a time interval of length t then has a Poisson distribution and the probability for k customers is

$$P(k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \qquad k = 0, 1, 2, ...$$

Compound Poisson process

The size of a customer demand is also a stochastic variable that is independent of the distribution of the customer arrivals. It is convenient to assume that each customer demands an integral number of units.

f_j = probability of demand size j (j = 1, 2, ...).

Poisson demand – forecasting

- Arrival rate and mean customer demand size: as discussed for forecasting

- Demand size distribution:
  - Either assume certain distribution type and estimate its parameter(s)
  - Or estimate directly from observed demand size frequencies (but periodic and not per customer)

Compound Poisson process

f_j = probability of demand size j (j = 1, 2, ...).

f_j^k = probability that k customers give the total demand j,

D(t) = stochastic demand in the time interval t.

$$f_0^0 = 1 \qquad f_j^1 = f_j \qquad f_j^k = \sum_{i=k-1}^{j-1} f_i^{k-1} f_{j-i}, \qquad k = 2, 3, 4, ...$$

$$P(D(t) = j) = \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} e^{-\lambda t} f_j^k$$

Discrete compound Poisson demand

To evaluate policies, the distribution of the inventory level (stock on hand - backorders) is needed.
We start by deriving the distribution of the inventory position (stock on hand + outstanding orders - backorders).

Proposition
If not all demands are multiples of some integer larger than one, then in steady state the inventory position is Discrete_uniformly distributed on R + 1, R + 2, ..., R + Q.

Proof of proposition (pure Poisson)

Periodic Markov chain

$$\begin{tikzcd}
R+Q \arrow[r, "\lambda"] & R+Q-1 \arrow[r, "\lambda"] & R+Q-2 & \cdots & R+2 \arrow[r, "\lambda"] & R+1 \arrow[lllll, bend left, "\lambda"]
\end{tikzcd}$$

Sketched proof of proposition (compound Poisson)

IF not all demands are multiples of some integer larger than one, we must reach all the considered inventory positions.

The jumps from R + i to R + j can be seen as a Markov chain.

This chain has a unique steady state distribution. It is easy to show that the uniform distribution is A solution and so THE solution.

Important observation

L = lead-time,
$IL(t)$ = inventory level (at time t),
$IP(t)$ = inventory position (at time t),
$D(t, t + \tau) = D(\tau)$ = stochastic demand in the interval $(t, t + \tau]$.

Orders that have been triggered in the interval $(t, t + L]$ have not reached the inventory due to the lead-time. Hence

$$\mathbf{IL(t + L) = IP(t) - D(t, t + L).}$$

Discrete compound Poisson demand

IL(t + L) = IP(t) - D(t, t + L).

$$P(IL = j) = \frac{1}{Q} \sum_{k=\max\{R+1, j\}}^{R+Q} P(D(L) = k - j) \quad j \le R + Q$$

Special case: if R = S - 1 and Q=1 then

$$P(IL = j) = P(D(L) = S - j) \quad j \le S.$$

Normal demand

Mean lead time demand

$$F(x) = P(IL \le x) = \frac{1}{Q}\int_{R}^{R+Q}\left[1-\Phi\left(\frac{u-x-\mu'}{\sigma'}\right)\right]du$$

Stand. dev. Lead time demand

We now introduce the loss function G(x)

$$G(x) = \int_x^\infty (v-x)\varphi(v)dv = \int_x^\infty v\varphi(v)dv - x(1-\Phi(x))$$

Using

$$\int_x^\infty v\varphi(v)dv = \int_x^\infty v\frac{1}{\sqrt{2\pi}}e^{-\frac{v^2}{2}}dv = \int_x^\infty \frac{1}{\sqrt{2\pi}}e^{-\frac{v^2}{2}}d\frac{v^2}{2}$$

$$= -\frac{1}{\sqrt{2\pi}}e^{-\frac{v^2}{2}}\Big|_x^\infty = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}} = \varphi(x)$$

we get $G(x) = \varphi(x) - x(1-\Phi(x))$

Excel: G(x) = NORMDIST(x;0;1;FALSE) - x*(1-NORMDIST(x;0;1;TRUE))

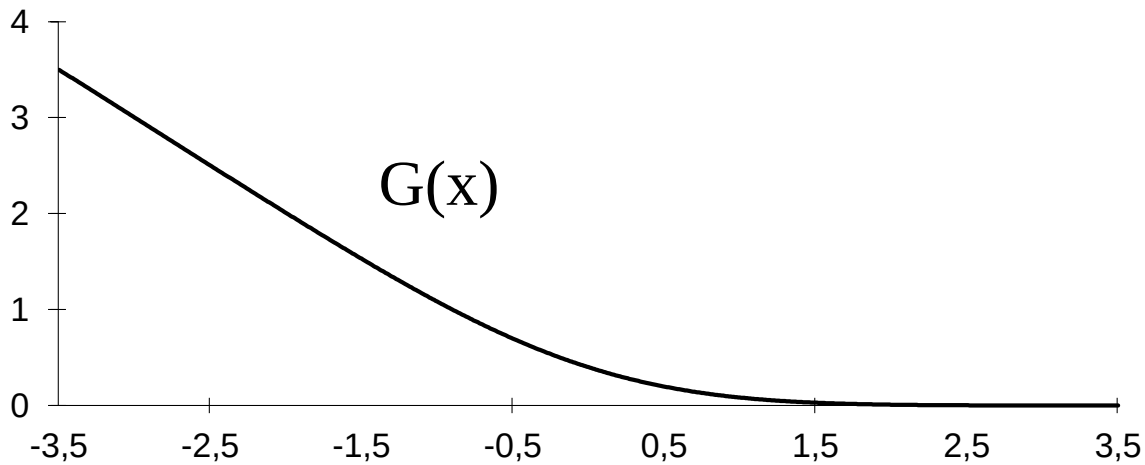
Normal demand

$$G(x) = \varphi(x) - x(1 - \Phi(x))$$

Note that $G'(x) = \varphi'(x) + x\varphi(x) - (1 - \Phi(x))$

$$= -x\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2} + x\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2} + (\Phi(x) - 1)$$

$$= \Phi(x) - 1$$



Normal demand (continued)

$$F(x) = P(IL \le x) = \frac{1}{Q}\int_{R}^{R+Q}\left[1-\Phi\left(\frac{u-x-\mu'}{\sigma'}\right)\right]du$$

$$G'(x) = \Phi(x) - 1,$$

$$F(x) = \frac{1}{Q}\int_{R}^{R+Q}\left[-G'\left(\frac{u-x-\mu'}{\sigma'}\right)\right]du = \frac{\sigma'}{Q}\left[G\left(\frac{R-x-\mu'}{\sigma'}\right) - G\left(\frac{R+Q-x-\mu'}{\sigma'}\right)\right]$$

$$f(x) = \frac{1}{Q}\int_{R}^{R+Q}\frac{1}{\sigma'}\varphi\left(\frac{u-x-\mu'}{\sigma'}\right)du = \frac{1}{Q}\left[\Phi\left(\frac{R+Q-x-\mu'}{\sigma'}\right) - \Phi\left(\frac{R-x-\mu'}{\sigma'}\right)\right]$$

Example (Normal versus Poisson)

Pure Poisson demand with $\lambda = 2$. Lead-time $L = 5$.
(R, Q) policy with $R = 9$ and $Q = 5$.
The lead-time demand is obviously Poisson (10) distributed.

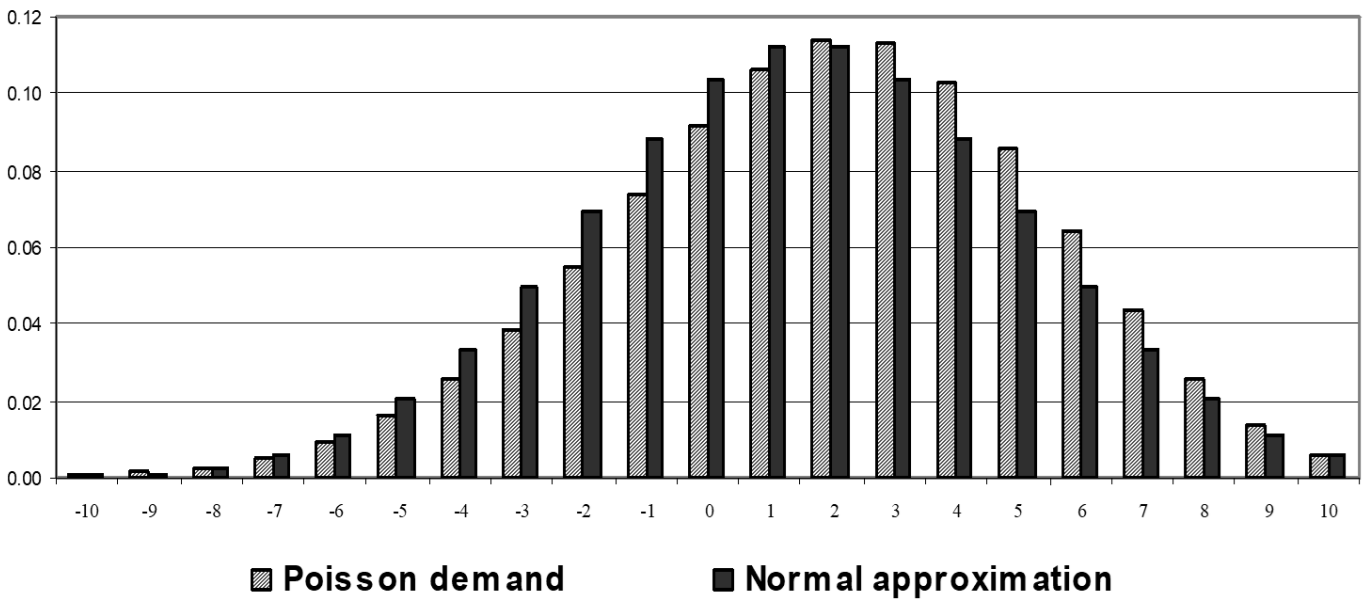
$$P(IL = j) = \frac{1}{5} \sum_{k=\max\{10,j\}}^{14} \frac{10^{k-j}}{(k-j)!} e^{-10}$$

Normal approximation with same mean $\mu' = 10$ and var. $\sigma' = 10^{1/2}$

$$F(x) = \frac{\sqrt{10}}{5}\left[G\left(\frac{-x-1}{\sqrt{10}}\right) - G\left(\frac{-x+4}{\sqrt{10}}\right)\right]$$

Example (continued)

Compare P(IL = j) with F(j + 0.5) - F(j - 0.5)



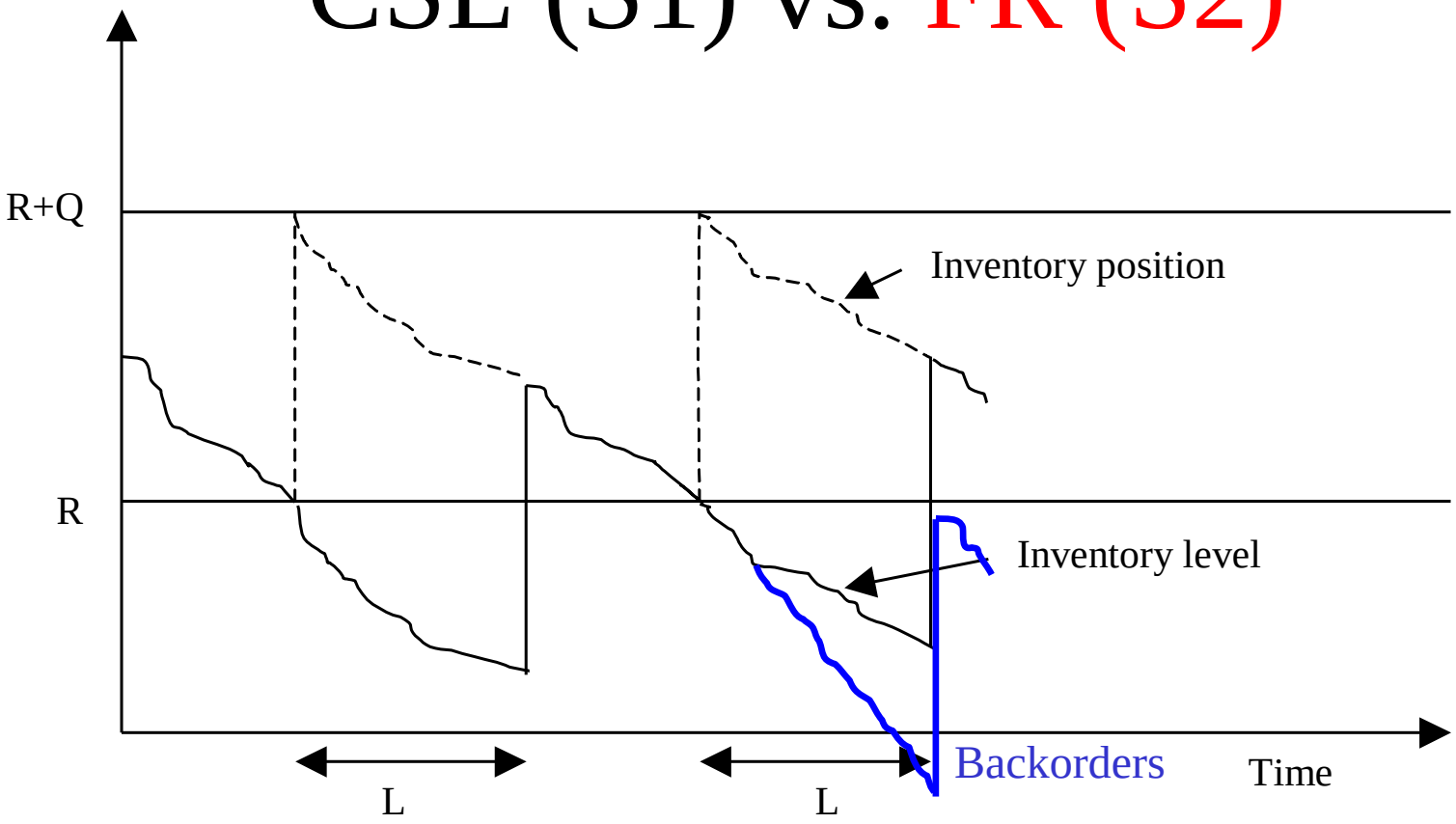
Service level definitions

S1 = “cycle service level” - probability of no stockout per order cycle

S2 = “fill rate” - fraction of demand that can be satisfied immediately from stock on hand

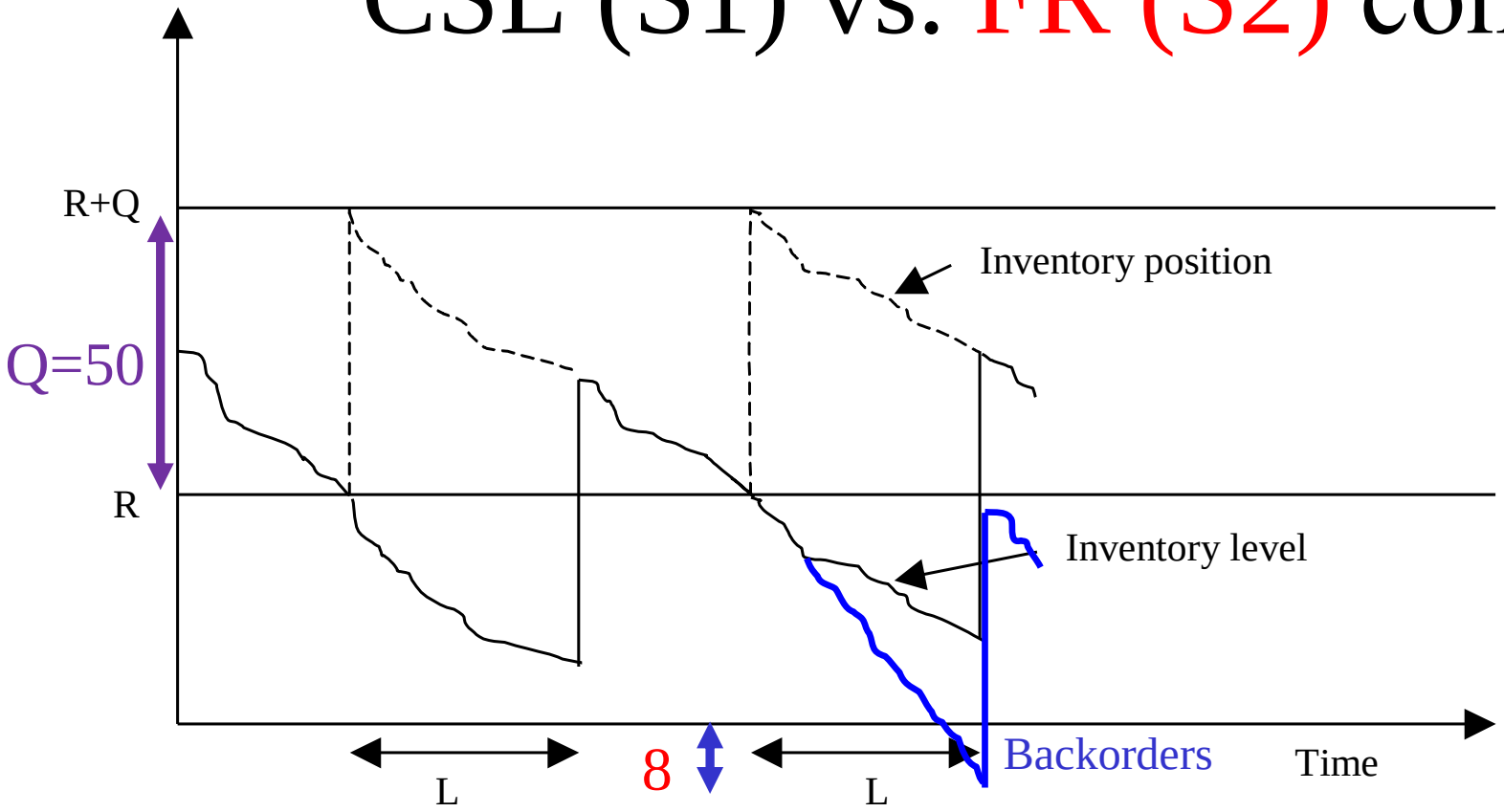
S3 = “ready rate” - fraction of time with positive stock on hand

CSL (S1) vs. FR (S2)



CSL: 50% of the orders arrived on time, but ...

CSL (S1) vs. FR (S2) cont'd



Out of 2×Q = 2×50 = 100 ordered SKUs, 8 satisfied Backorders and so 92% demands where satisfied directly from stock on hand.

Determining the safety stock for given cycle service level S_1

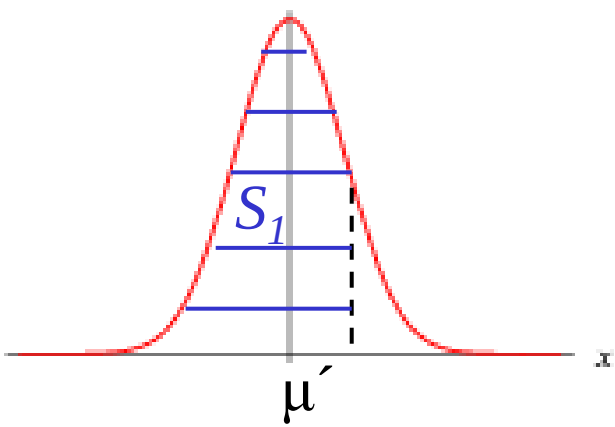
$$P(D(L) \le R) = S_1 = \Phi\left(\frac{R - \mu'}{\sigma'}\right) = \Phi\left(\frac{SS}{\sigma'}\right)$$

$$SS = k\sigma' \quad \text{with} \quad k = \Phi^{-1}(S_1) \qquad R = SS + \mu'.$$

$$\text{Holding cost} \approx h\left(R + \frac{Q}{2} - \mu'\right) = h\left(SS + \frac{Q}{2}\right)$$

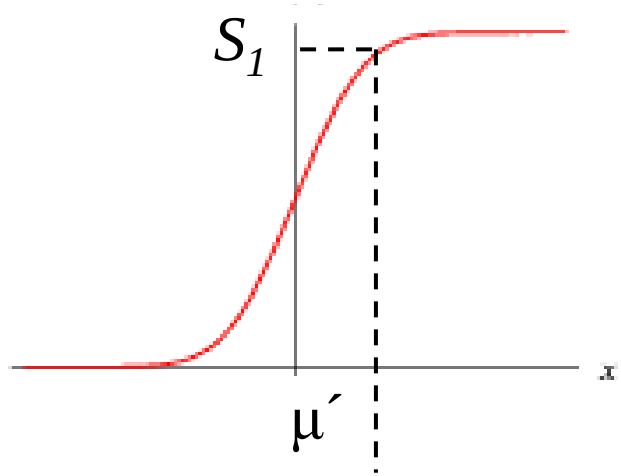
Normal distribution

Probability density function



Area is S_1

Probability distribution function

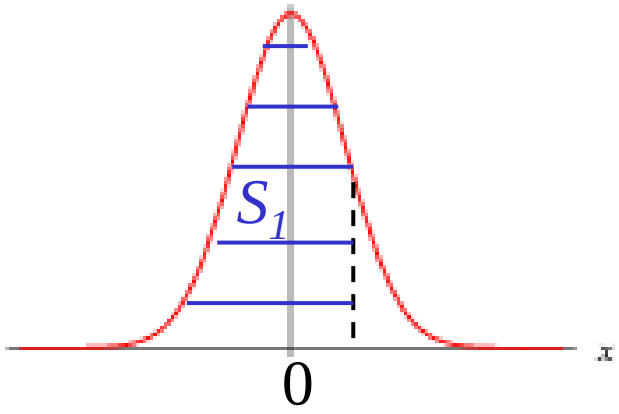


Excel!!

$$R = \text{Inverse Normal}\,(\mu', \sigma')\ \text{pdf at}\ S_1$$

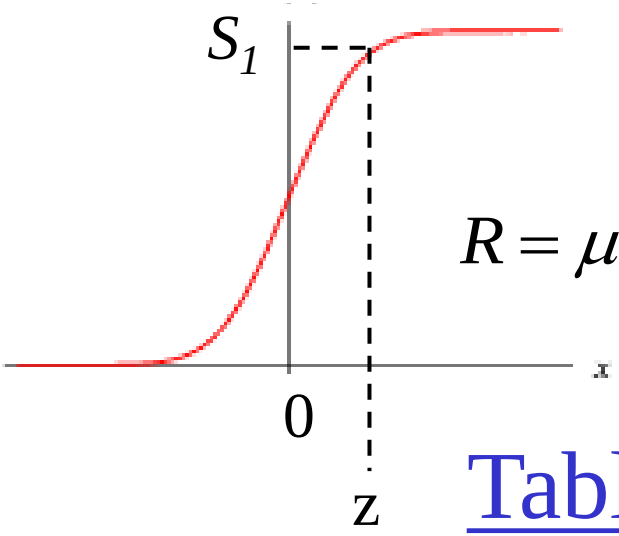
Standard Normal distribution

Probability density function



Area is S_1

Probability distribution function



$$R = \mu' + k\sigma'$$

Table

Cycle service level – safety factor

Safety factors for different values of service level S_1.

Service level S_1	0.75	0.80	0.85	0.90	0.95	0.99
Safety factor k	0.67	0.84	1.04	1.28	1.64	2.33

Fill rate S_2 & Ready rate S_3

In general, the fill rate is not equal to the ready rate.

Example. Demands are always of size 1 and the time between demands is always 7 days. An (R,Q) policy with R=0 and Q=1 is used. The lead time is 3 days.

Fill rate S_2 & Ready rate S_3

The fill rate is equal to the ready rate for

- Unit Poisson demand (*PASTA property*)
- Continuous (e.g. Normal) demand

S_2 & S_3: Compound Poisson demand

$$S_2 = \frac{\sum_{k=1}^{\infty}\sum_{j=1}^{\infty} \min(j,k) \cdot f_k \cdot P(IL = j)}{\sum_{k=1}^{\infty} k\, f_k}$$

f_k: probability that a demand is of size k

$$S_3 = P(IL > 0)$$

S_2 & S_3: Normal demand

$$S_2 = S_3 = 1 - F(0) = 1 - \frac{\sigma'}{Q}\left[G\left(\frac{R-\mu'}{\sigma'}\right) - G\left(\frac{R+Q-\mu'}{\sigma'}\right)\right]$$

$$S_2 = S_3 \approx 1 - \frac{\sigma'}{Q} G\left(\frac{R-\mu'}{\sigma'}\right)$$

S_2 & S_3: Example

Pure Poisson demand with demand rate $\lambda = 2$, lead time $L = 5$, continuous review (R, Q) policy with $R = 9$ and $Q = 5$.

Exact $$S_2 = S_3 = P(IL > 0) = \sum_{j=1}^{14} \frac{1}{5} \sum_{k=\max\{10,j\}}^{14} \frac{10^{k-j}}{(k-j)!} e^{-10} = 0.679$$

Normal approx. $$S_2 = S_3 = 1 - F(0) = 1 - \frac{\sqrt{10}}{5}\left[G\left(\frac{-1}{\sqrt{10}}\right) - G\left(\frac{4}{\sqrt{10}}\right)\right] = 0.666$$

FR to R approx. – Normal demand

Standardized(loss) per cycle: $$I = \frac{Q(1 - S_2)}{\sigma'}$$

Using Equation (12a) in Shore (1982)

$$S_1 \approx \begin{cases} \dfrac{e^{1-I/0.4115}}{1 + e^{1-I/0.4115}} & \text{if } I > 0.4115 \\ \dfrac{0.4115}{0.4115 + I} & \text{if } I \leq 0.4115 \end{cases}$$

$$R = \mu' + \sigma' \Phi^{-1}(S_1)$$

Backorder (shortage) costs (alternative for service level)

b1 = shortage cost per unit and time unit,

b2 = shortage cost per unit.

Backorder cost b_1 per unit (continuous Normal demand)

$$C = hE\left[IL^{+}\right] + b_1 E\left[IL^{-}\right]$$
$$= hE\left[IL\right] + \left(h + b_1\right)E\left[IL^{-}\right]$$
$$= h(R + Q/2 - \mu') + \left(h + b_1\right)\int_{-\infty}^{0} F(x)dx$$

Recall:

$$F(x) = P(IL \le x) = \frac{1}{Q}\int_{R}^{R+Q}\left[-G'\left(\frac{u - x - \mu'}{\sigma'}\right)\right]du = \frac{\sigma'}{Q}\left[G\left(\frac{R - x - \mu'}{\sigma'}\right) - G\left(\frac{R + Q - x - \mu'}{\sigma'}\right)\right]$$

$$\int_{-\infty}^{0} G\left(\frac{R - x - \mu'}{\sigma'}\right) dx = \cdots$$

$$v = \frac{R - x - \mu'}{\sigma'} \quad \text{and so} \quad \frac{dx}{dv} = -\sigma'$$

$$\ldots = \int_{\infty}^{\frac{R-\mu'}{\sigma'}} G(v) \frac{dx}{dv} dv = (- -)\sigma' \int_{\frac{R-\mu'}{\sigma'}}^{\infty} G(v)\, dv$$

$$F(x) = P(IL \le x) = \frac{1}{Q}\int_{R}^{R+Q}\left[-G'\left(\frac{u-x-\mu'}{\sigma'}\right)\right]du = \frac{\sigma'}{Q}\left[G\left(\frac{R-x-\mu'}{\sigma'}\right) - G\left(\frac{R+Q-x-\mu'}{\sigma'}\right)\right]$$

$$\int_{-\infty}^{0} F(x)\,dx = \frac{\sigma'}{Q}\int_{-\infty}^{0}\left[G\left(\frac{R-x-\mu'}{\sigma'}\right) - G\left(\frac{R+Q-x-\mu'}{\sigma'}\right)\right]dx$$

$$= \frac{\sigma'}{Q}\left[\sigma'\int_{\frac{R-\mu'}{\sigma'}}^{\infty} G(v)\,dv - \sigma'\int_{\frac{R+Q-\mu'}{\sigma'}}^{\infty} G(v)\,dv\right]$$

$$= \frac{\sigma'^{2}}{Q}\left[H\left(\frac{R-\mu'}{\sigma'}\right) - H\left(\frac{R+Q-\mu'}{\sigma'}\right)\right]$$

where $\mathrm{H(x)} = \int_{\mathrm{x}}^{\infty}\mathrm{G(v)dv} = \frac{1}{2}\left[(\mathrm{x}^{2}+1)(1-\Phi(\mathrm{x})) - \mathrm{x}\varphi(\mathrm{x})\right]$

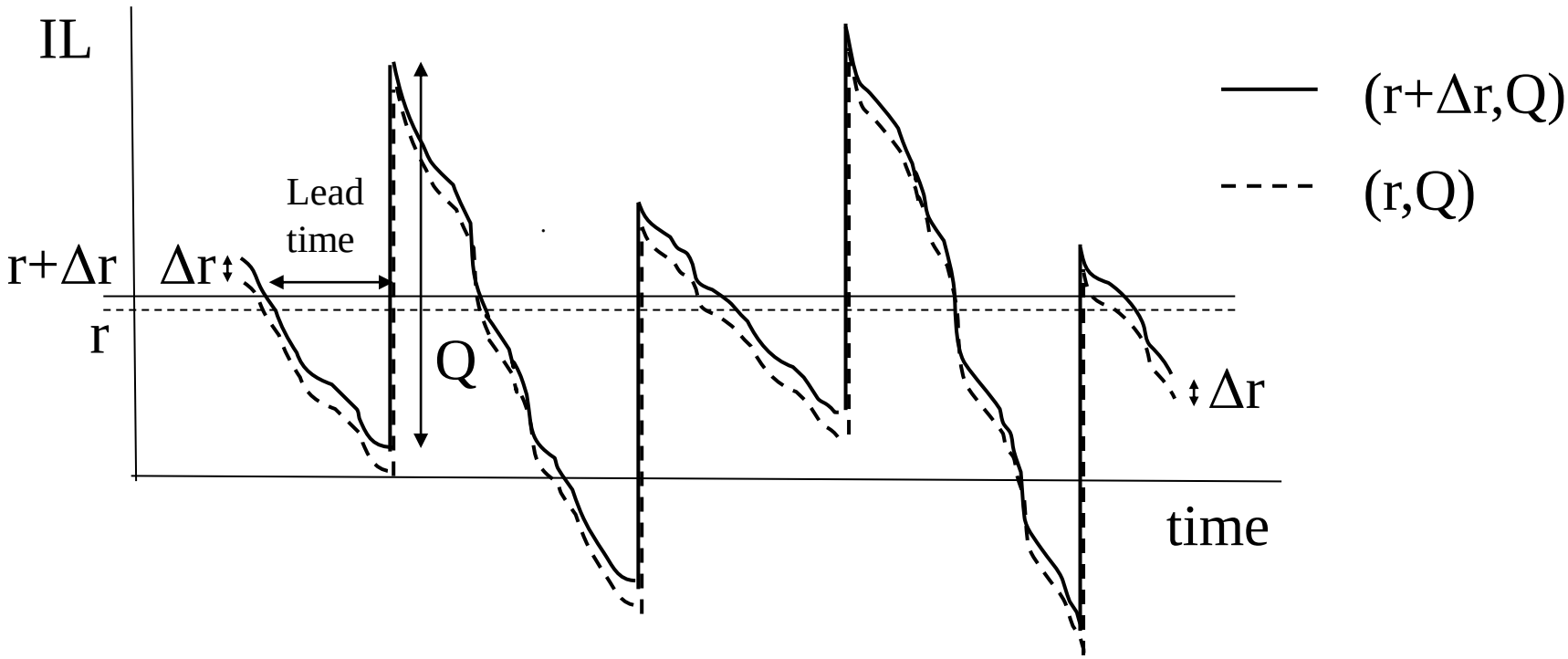
$$C = hE\left[IL^{+}\right] + b_1 E\left[IL^{-}\right]$$
$$= hE\left[IL\right] + \left(h + b_1\right)E\left[IL^{-}\right]$$
$$= h(R + Q/2 - \mu') + \left(h + b_1\right)\int_{-\infty}^{0} F(x)dx$$

$$\mathrm{C} = \mathrm{h}(\mathrm{R} + \mathrm{Q}/2 - \mu') + (\mathrm{h} + \mathrm{b}_1)\frac{\sigma'^2}{\mathrm{Q}}\left[\mathrm{H}\left(\frac{\mathrm{R} - \mu'}{\sigma'}\right) - \mathrm{H}\left(\frac{\mathrm{R} + \mathrm{Q} - \mu'}{\sigma'}\right)\right]$$

where $\mathrm{H}(\mathrm{x}) = \int_{\mathrm{x}}^{\infty} \mathrm{G}(\mathrm{v})\mathrm{dv} = \frac{1}{2}\left[(\mathrm{x}^2 + 1)(1 - \Phi(\mathrm{x})) - \mathrm{x}\varphi(\mathrm{x})\right]$

FOC leads to optimality condition, but marginal cost analysis on the next slide is easier and more insightful (and applies also to non-Normal demand)

Backorder cost b_1 per unit and time unit (General demand)



$$C(r+\Delta r)-C(r)=h\Delta r S_3(r)-b_1\Delta r\left(1-S_3(r)\right)+o(\Delta r)$$

In the optimal solution we have

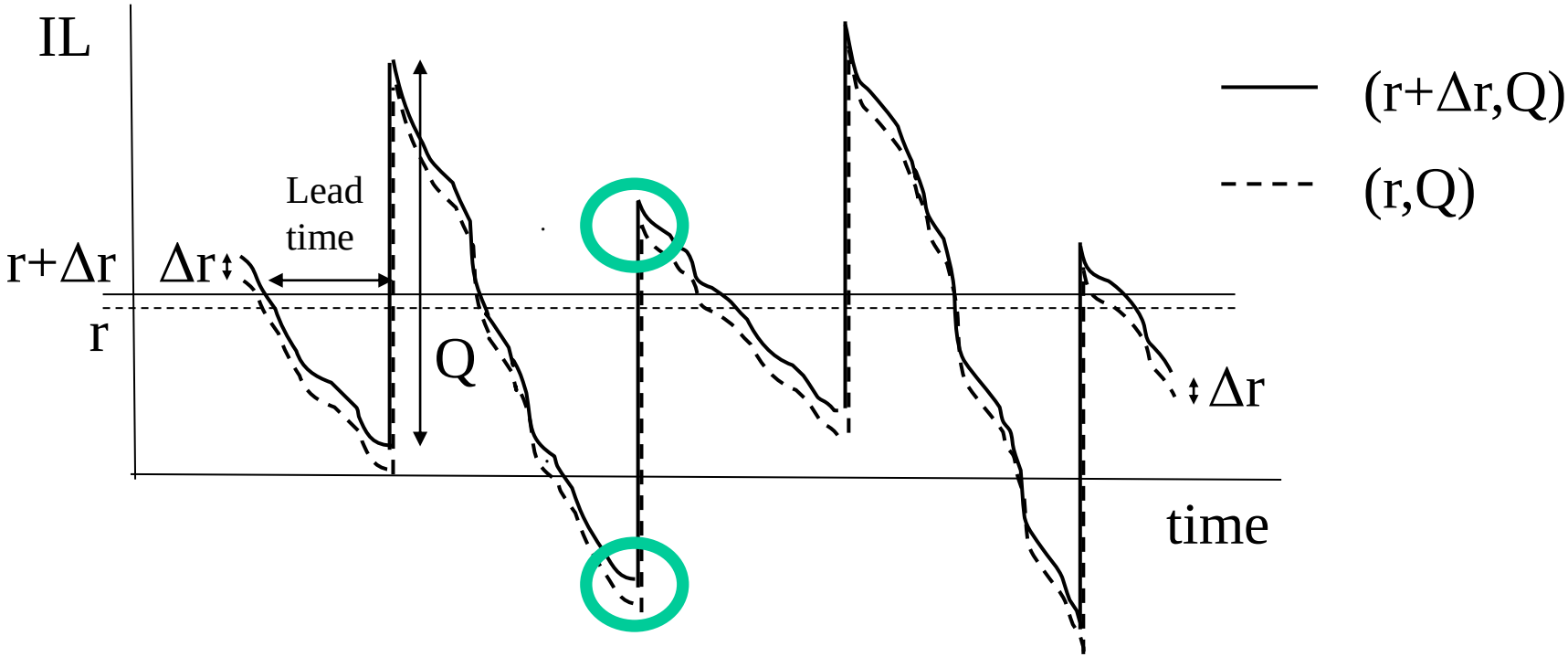
$$S_2=S_3=\frac{b_1}{h+b_1}$$

Backorder cost b_2 per unit (continuous Normal demand)

$$F(x) = P(IL \le x) = \frac{1}{Q}\int_{R}^{R+Q}\left[-G'\left(\frac{u-x-\mu'}{\sigma'}\right)\right]du = \frac{\sigma'}{Q}\left[G\left(\frac{R-x-\mu'}{\sigma'}\right) - G\left(\frac{R+Q-x-\mu'}{\sigma'}\right)\right]$$

$$C = h(R+Q/2-\mu') + h\int_{-\infty}^{0} F(x)dx + b_2\mu F(0)$$

Backorder cost b_2 per unit (General demand)



$$C(r+\Delta r) - C(r)$$
$$= h\Delta r S_3(r) - b_2 \frac{\mu}{Q}\Delta r\left(\left(1 - S_1(r)\right) - \left(1 - S_1(r+Q)\right)\right) + o(\Delta r)$$

In the optimal solution we have approximately (Q large)

$$S_1 \approx 1 - \frac{QhS_3}{b_2\mu}$$

Optimality of ordering policies

For most inventory problems with a single-echelon, independent items, and stationary stochastic demand the optimal policy is indeed of the (s, S) type.

Given a fixed batch quantity, an (R, Q) policy is optimal.

Lost Sales

It no longer holds that

$$IL(t+L) = IP(t) - D(t, t+L).$$

An exact analysis is therefore more complex.

(R,Q) or (s,S) policies are generally no longer optimal. They are still a reasonable choice, though, because the fraction of demand lost is usually small.

What about forecasting?

- So far, we have assumed that the (lead time) demand mean and variance are given.

- In practice, these need to be forecasted!

- Indeed, we provided methods for estimating the per period demand mean μ and standard deviation σ in the forecasting lecture.
- But how to convert these to lead time demand estimates?

Dennis Prak, Aris Syntetos, Ruud Teunter, On the Calculation of Safety Stocks when Demand is Forecasted, *European Journal of Operational Research* 256(2), 454–461, 2017.

Forecast-based SS (L=1)

- Theory: given expected demand per period μ
- $E[x_{t+1} - \mu]^2 = V[x_{t+1} - \mu] = \sigma^2$

- Practice: given forecast $\hat{x}_{t,t+1}$
- For unbiased estimators we get
- $E[x_{t+1} - \hat{x}_{t,t+1}]^2 = V[x_{t+1} - \hat{x}_{t,t+1}]$
- And under $Cov[x_{t+1}, \hat{x}_{t,t+1}]=0$ this gives
- $E[x_{t+1} - \hat{x}_{t,t+1}]^2 = V[x_{t+1}] + V[\hat{x}_{t,t_1}] = \sigma^2 + V[\hat{x}_{t,t+1}]$

MA(n) forecast (L=1)

- Assume <u>stationary</u>, normal demand
- $\hat{x}_t \equiv \hat{x}_{t,t+k}$
- Use MA(n): $\hat{x}_t = (x_{t-n+1} + \ldots + x_t)/n$
- Note: unbiased and zero covariance

- $E[x_{t+1} - \hat{x}_t]^2 = V[x_{t+1}] + V[\hat{x}_t] = \sigma^2 + \sigma^2/n$
- Good news: This is indeed equal to the expected MSE of the one step ahead forecast

Forecast-based SS (L>1)

- Lead time demand forecast error
- $\sum_{k=1}^{L}(x_{t+k} - \hat{x}_t) = \sum_{k=1}^{L} x_{t+k} - L\hat{x}_t$

- The variance is
- $\sum_{k=1}^{L} V[x_{t+k}] - L^2 V[\hat{x}_t]$
- (under stationary demand, unbiased forecast)

MA(n) forecast (L>1)

- Assume stationary, normal demand
- Use MA(n)

- $L \times V[x_{t+1}] + L^2 \times V[\hat{x}_t]$

$= L\sigma^2 + L^2 \times \sigma^2/n$

$> L\sigma^2 + L \times \sigma^2/n$ (estimated by L time the MSE of the one step ahead forecast)

Increase in SS needed

	Increase in safety stock for			
L	$N = 1$	$N = 4$	$N = 12$	$N = 52$
1	0%	0%	0%	0%
2	22%	10%	4%	1%
3	41%	18%	7%	2%
4	58%	26%	11%	3%
5	73%	34%	14%	4%
6	87%	41%	18%	5%

But σ is also unknown …

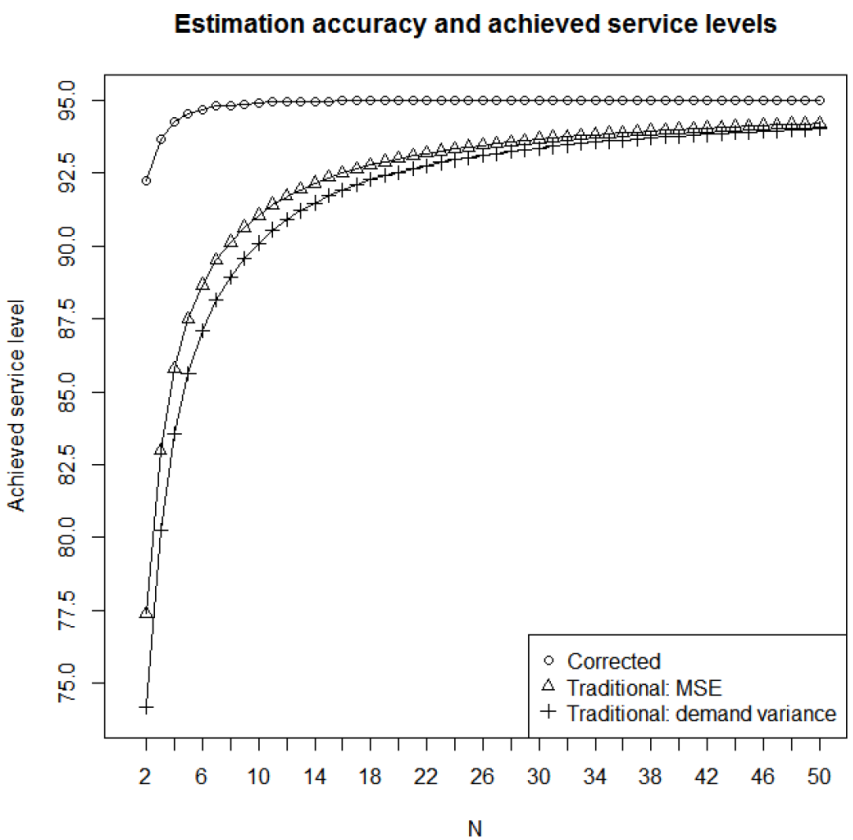


Figure 1: Number of demand observations (N) and achieved service levels (γ') for $\mu = 10$, $\sigma = 2$, $L = 4$, and $\gamma^* = 95\%$.

Dennis Prak, Aris Syntetos, Ruud Teunter, On the Calculation of Safety Stocks when Demand is Forecasted, *European Journal of Operational Research* 256(2), 454–461, 2017.